

The inter-region inequality at $m = 2$ is false,
and condition (A) at $m = 2$ is true

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Abstract

The brief for this session asked for a proof of the inter-region inequality (G), the single remaining gap in 2026-09-30-c1-cylindric-kostka-logconcavity.tex. Step 0 of the brief — extend the verification range before proving anything — refutes it. **(G) is false:** at $n = 10$, $\mu = (0, 5)$, $\lambda = (5, 12)$, $b = 4$ the region partial sums $G_I = (1, 3, 6, 7, 6, 3, 1)$ and $G_{III} = (1, 1, 1, 1, 1, 1, 1)$ share the support $[10, 16]$ and the pairwise inequality reads $6 \geq 7$ at $s = 11$. The failure is structural, not marginal: against a partner that is *constant* across one's support, the pairwise inequality (*) is *literally concavity*, and concavity of the region sums was already proved false four pages later in the same paper. The route was refuted by a fact the paper itself contains.

Chasing the mechanism, however, produced the theorem. Two one-line identities,

$$a_t + d_t = t + D, \quad b_t + c_t = u + B - t,$$

force *every* trapezoid in the slice sum to be symmetric about the *same* centre $\sigma_0 = \frac{1}{2}(u + \lambda_1 + \lambda_2)$. Hence G is a radial function, $G(s) = \gamma(|s - \sigma_0|)$, where γ is the upper tail of a sequence β that is the positive part of a concave function. Since the upper tail of any PF₂ sequence is log-concave, G is log-concave. This proves **condition (A) at $m = 2$ outright** — all slices, not 79.9% — without (G), and it explains in one identity why (G) and concavity both fail: $2\gamma(y) - \gamma(y - 1) - \gamma(y + 1) = \beta(y) - \beta(y - 1)$.

1 What is proved and what is refuted

Throughout, $m = 2$, $\mu = (\mu_1, \mu_2)$, $\lambda = (\lambda_1, \lambda_2)$, $n \geq 2$ is the level, $d = |\lambda/\mu|$, $\ell = 3$, and $k(a, b) = K_{(a, b, d - a - b)}^c$ as in 2026-09-30-c1-cylindric-kostka-logconcavity.tex (henceforth [c1]). Write PF₂ for: nonnegative, support an interval of $\mathbb{Z}$, and $g(s)^2 \geq g(s - 1)g(s + 1)$ for all s .

Theorem 1.1 (Condition (A) at $m = 2$). *For every cylindric skew shape λ/μ with $m = 2$, every level $n \geq 2$ and every b , the sequence $a \mapsto k(a, b)$ is PF₂. In particular $k(a, b)^2 \geq k(a + 1, b)k(a - 1, b)$ for all a, b : condition (A) of [c1, Conj. 2.1] holds at $m = 2$.*

Proposition 1.2 (the inter-region inequality is false). *Inequality (G) of [c1, §"The gap"] — that*

$$2G_R(s)G_{R'}(s) \geq G_R(s - 1)G_{R'}(s + 1) + G_R(s + 1)G_{R'}(s - 1) \quad \text{for all } s \quad (\text{G})$$

for every pair of distinct regions — is false. Over $2 \leq n \leq 11$, $m = 2$, $d \leq 14$ (201 147 nonempty slices, 37 672 occurring region pairs) it fails on exactly 92 pairs; every failure is the pair I–III; and every failure has all three of I, II/IV, III nonempty.

Theorem 1.1 does not go through (G): it replaces the whole region decomposition. Section 2 records the refutation and its mechanism; §3–§6 prove Theorem 1.1; §7 shows that the mechanism of the refutation is an *identity* in the new picture, so (G) is dead with no residual scope.

2 The refutation

2.1 Setting, recalled so that nothing is reconstructed from memory

Put $A = \lambda_2 - n + 1$, $B = \lambda_1$, $C = \lambda_1 + 1$, $D = \lambda_2$, so that

$$C = B + 1, \quad A = D - n + 1. \quad (1)$$

Fix b and set $u = |\mu| + b$. By [c1, Prop. 2.3] (**A-slice-sum-reformulation, proved**), $k(a, b) = G(u + a)$ where

$$G(s) = \sum_{t \in T} (\mathbf{1}_{[a_t, b_t]} * \mathbf{1}_{[c_t, d_t]})(s), \quad T = [T_-, T_+], \quad (2)$$

$$a_t = \max(t, A), \quad b_t = \min(u - 1 - t, B), \quad c_t = \max(u - t, C), \quad d_t = \min(t + n - 1, D), \quad (3)$$

and $T_- = \max(\mu_1, u - \mu_1 - n + 1)$, $T_+ = \min(\mu_2 - 1, u - \mu_2)$. Here $t = \nu_1$ indexes the first interior layer of the chain, $\nu_2 = u - t$, and $s = \kappa_1 + \kappa_2$. Values of t for which $a_t > b_t$ or $c_t > d_t$ contribute 0. Write

$$n_t = b_t - a_t + 1, \quad m_t = d_t - c_t + 1, \quad h_t = \min(n_t, m_t).$$

The two breakpoints are $\tau_1 = A$ (for a_t and d_t) and $\tau_2 = u - 1 - B = u - C$ (for b_t and c_t), and the four regions of [c1] are indexed by $(\mathbf{1}[t \geq \tau_1], \mathbf{1}[t > \tau_2]) = (0, 0), (1, 0), (1, 1), (0, 1)$ for I, II, III, IV respectively, with G_R the corresponding partial sum of (2).

2.2 Only three regions ever occur, and they are a chain

Lemma 2.1. T_{II} and T_{IV} are never both nonempty. Consequently at most three regions occur, and in increasing order of t they are I, II, III (if $\tau_1 \leq \tau_2$) or I, IV, III (if $\tau_2 < \tau_1$).

Proof. $T_{\text{II}} = \{t \in T : \tau_1 \leq t \leq \tau_2\} \neq \emptyset$ forces $\tau_1 \leq \tau_2$, while $T_{\text{IV}} = \{t \in T : \tau_2 < t < \tau_1\} \neq \emptyset$ forces $\tau_2 < \tau_1$. The order statement is immediate from the defining inequalities. $\square$

This already corrects the brief, which proposed attacking the pair II–IV first on the ground that both are $\rho \circ H$ with a shared H . That pair is *vacuous*; consistently, the census of [c1] lists II–IV nowhere and its five pair counts sum to its total ($177 + 177 + 81 + 45 + 45 = 525$).

2.3 The mechanism: against a flat partner, (*) is concavity

Lemma 2.2. Let $g, h \geq 0$ and suppose $h \equiv c > 0$ on an interval J . If $s - 1, s, s + 1 \in J$ then

$$2g(s)h(s) \geq g(s - 1)h(s + 1) + g(s + 1)h(s - 1) \iff 2g(s) \geq g(s - 1) + g(s + 1).$$

Proof. Both sides of the left inequality equal c times the corresponding side of the right one. $\square$

So whenever one region sum is constant and positive across a window containing three consecutive points of another's support, (G) for that pair *is* concavity of the other at those points. And [c1, §"Concavity is false"] already proves concavity of the slice sums false, with witness $k(\cdot, 2) = (1, 3, 6, 7, 6, 3, 1)$ at $n = 8$, $\mu = (0, 3)$, $\lambda = (4, 7)$. All that is needed to refute (G) is a shape in which a non-concave region sum coexists with a *flat* one on the same support. It exists.

2.4 The witness

Proposition 2.3. *Let $n = 10$, $\mu = (0, 5)$, $\lambda = (5, 12)$, $b = 4$ (so $d = 12$, $u = 9$, $A = 3$, $B = 5$, $C = 6$, $D = 12$, $\tau_1 = \tau_2 = 3$, $T = [0, 4]$). Then*

t	$[a_t, b_t] \times [c_t, d_t]$	region	trapezoid on its support
0	$[3, 5] \times [9, 9]$	I	$(1, 1, 1)$ on $[12, 14]$
1	$[3, 5] \times [8, 10]$	I	$(1, 2, 3, 2, 1)$ on $[11, 15]$
2	$[3, 5] \times [7, 11]$	I	$(1, 2, 3, 3, 3, 2, 1)$ on $[10, 16]$
3	$[3, 5] \times [6, 12]$	II	$(1, 2, 3, 3, 3, 3, 3, 2, 1)$ on $[9, 17]$
4	$[4, 4] \times [6, 12]$	III	$(1, 1, 1, 1, 1, 1, 1)$ on $[10, 16]$

so that, all on $[10, 16]$,

$$G_I = (1, 3, 6, 7, 6, 3, 1), \quad G_{III} = (1, 1, 1, 1, 1, 1, 1),$$

and $G_{II} = (1, 2, 3, 3, 3, 3, 3, 2, 1)$ on $[9, 17]$, $G_{IV} = 0$. Each G_R is PF₂. At $s = 11$,

$$2G_I(11)G_{III}(11) = 2 \cdot 3 \cdot 1 = 6 < 7 = 1 \cdot 1 + 6 \cdot 1 = G_I(10)G_{III}(12) + G_I(12)G_{III}(10),$$

so (G) fails for the pair I–III (and again at $s = 15$, by symmetry). Nevertheless $G = (1, 4, 7, 10, 11, 10, 7, 4, 1)$ on $[9, 17]$ is PF₂ — indeed concave on its support.

Proof. Direct substitution in (3); the region labels are read off from $\mathbf{1}[t \geq 3]$, $\mathbf{1}[t > 3]$. G_I is the sum of the first three trapezoids, $(1, 1, 1)$ on $[12, 14]$ plus $(1, 2, 3, 2, 1)$ on $[11, 15]$ plus $(1, 2, 3, 3, 3, 2, 1)$ on $[10, 16]$, which is $(1, 3, 6, 7, 6, 3, 1)$ on $[10, 16]$. The displayed numerical inequality is $6 < 7$. The region III sum is a single trapezoid with $n_4 = 1$, hence an interval indicator, hence constant on $[10, 16]$; so by Lemma 2.2 the failure at $s = 11$ is exactly the failure of concavity of G_I there, $2 \cdot 3 < 1 + 6$. $\square$

Observe that $(1, 3, 6, 7, 6, 3, 1)$ is the *same* sequence that refutes concavity in [c1, §”Concavity is false”] and in the registry node **A-concavity**; it reappears here as a region partial sum rather than as a full slice sum. The 525/525 sweep of [c1] ($n \leq 8$, $d \leq 10$) missed the refutation because no slice in that range places a non-concave region sum beside a flat one; the smallest witnesses have $n = 10$, $d = 12$.

Remark 2.4 (census). Over $2 \leq n \leq 11$, $m = 2$, $d \leq 14$: 201 147 nonempty slices; 111 009 meet one region, 30 181 meet two, 2 497 meet three (and 57 460 have $G \equiv 0$); 37 672 occurring region pairs, of which II–IV accounts for 0, confirming Lemma 2.1. (G) fails on 92 pairs, all I–III, all with three regions nonempty. In every one of the 92 failures the total G is still PF₂, and condition (A) fails on *none* of the 201 147 slices. In all 92 the mechanism of Lemma 2.2 applies: one of the two sums is constant across the support of the other.

Corollary 2.5. *The pairwise route of [c1, eq. (*)] cannot close condition (A) at $m = 2$. Since (*) also fails between individual slice members ([c1, §2.1 item 1]: $n = 6$, $\mu = (0, 3)$, $\lambda = (4, 5)$), the symmetrisation identity is unusable at both available granularities — members and regions.*

3 Two identities, and concentricity

The refutation says the region decomposition is the wrong decomposition. The following two identities say what the right one is. They are the entire content of the proof: everything afterwards is bookkeeping.

Lemma 3.1 (crossed sums). *For every $t \in \mathbb{Z}$,*

$$a_t + d_t = t + D, \quad b_t + c_t = u + B - t.$$

Proof. By (1), $a_t + d_t = \max(t, A) + \min(t + n - 1, A + n - 1)$. If $t \leq A$ this is $A + (t + n - 1)$; if $t \geq A$ it is $t + (A + n - 1)$. Both equal $t + A + n - 1 = t + D$.

Again by (1), $b_t + c_t = \min(u - 1 - t, B) + \max(u - t, B + 1)$. If $u - 1 - t \leq B$ then $u - t \leq B + 1$ and the value is $(u - 1 - t) + (B + 1)$; if $u - 1 - t \geq B$ then $u - t \geq B + 1$ and the value is $B + (u - t)$. Both equal $u + B - t$. $\square$

The two hypotheses $A = D - n + 1$ and $C = B + 1$ are each used exactly once, and each is used to make a max/min pair agree *on both branches* rather than merely at the breakpoint. This is strictly stronger than the statement in [c1] that the four endpoint functions share only two breakpoints: it says the two *crossed* sums are affine in t with slopes $+1$ and -1 globally.

Proposition 3.2 (concentricity). *Put*

$$\sigma_0 = \frac{u + B + D}{2} = \frac{u + \lambda_1 + \lambda_2}{2}, \quad t_0 = \frac{u + B - D}{2}.$$

Then for every $t \in \mathbb{Z}$,

$$(a_t + c_t) + (b_t + d_t) = 2\sigma_0, \tag{4}$$

$$n_t - m_t = u + B - D - 2t = 2(t_0 - t). \tag{5}$$

*Consequently every trapezoid $\text{Trap}_t := \mathbf{1}_{[a_t, b_t]} * \mathbf{1}_{[c_t, d_t]}$ in (2) is symmetric about the same point σ_0 , and G is symmetric about σ_0 .*

Proof. $(a_t + c_t) + (b_t + d_t) = (a_t + d_t) + (b_t + c_t) = (t + D) + (u + B - t) = u + B + D$ by Lemma 3.1, giving (4). And $n_t - m_t = (b_t - a_t) - (d_t - c_t) = (b_t + c_t) - (a_t + d_t) = (u + B - t) - (t + D)$, giving (5).

Trap_t is supported on $[a_t + c_t, b_t + d_t]$ and, being a convolution of two interval indicators, is symmetric about the midpoint of that interval, which by (4) is σ_0 — independently of t . A sum of functions symmetric about a common point is symmetric about it. $\square$

Note that σ_0 and t_0 do not depend on t , and that $\sigma_0 - t_0 = D$, $\sigma_0 + t_0 = u + B$. Set

$$L_t := \frac{n_t + m_t - 2}{2}, \quad \delta_t := \frac{|n_t - m_t|}{2} \stackrel{(5)}{=} |t - t_0|,$$

so that Trap_t is supported on $[\sigma_0 - L_t, \sigma_0 + L_t]$ and $L_t - \delta_t = h_t - 1$, $L_t + \delta_t = \max(n_t, m_t) - 1$. Both L_t and δ_t lie in $\mathbb{Z} - t_0$; indeed $L_t = n_t + (t - t_0) - 1$ by (5).

Lemma 3.3 (the tent). *$t \mapsto L_t$ is concave on $\mathbb{Z}$ with increments in $\{-1, 0, 1\}$; its increments are $+1$ for $t < \min(\tau_1, \tau_2)$, 0 on $[\min(\tau_1, \tau_2), \max(\tau_1, \tau_2)]$, and -1 for $t \geq \max(\tau_1, \tau_2)$. Writing $\tau_\wedge = \min(\tau_1, \tau_2)$, $\tau_\vee = \max(\tau_1, \tau_2)$ and $\Lambda = L_{\tau_\wedge} = L_{\tau_\vee}$, we have for all $t \in \mathbb{R}$*

$$\tilde{L}(t) = \min(t - \alpha, \Lambda, \omega - t), \quad \alpha = \tau_\wedge - \Lambda, \quad \omega = \tau_\vee + \Lambda, \tag{6}$$

where $\tilde{L}$ is the piecewise affine extension of L , and

$$\Lambda = \sigma_0 - \max(u, A + C). \tag{7}$$

Proof. $n_t = \min(u - 1 - t, B) - \max(t, A) + 1$ is a concave function of t (a concave minus a convex one) whose increment is $-\mathbf{1}[t \geq \tau_2] - \mathbf{1}[t \geq \tau_1]$ (each min/max contributing -1 past its own breakpoint); symmetrically m_t is concave with increment $\mathbf{1}[t < \tau_1] + \mathbf{1}[t < \tau_2]$. Hence $n_t + m_t$ is concave with increment

$$(\mathbf{1}[t < \tau_1] - \mathbf{1}[t \geq \tau_1]) + (\mathbf{1}[t < \tau_2] - \mathbf{1}[t \geq \tau_2]) \in \{-2, 0, 2\},$$

equal to $+2$ iff $t < \tau_\wedge$ and to -2 iff $t \geq \tau_\vee$. Dividing by 2 gives the statement about L , and a concave piecewise affine function with increments $+1, 0, -1$ in that order is exactly (6).

For (7): by (4), $L_t = \sigma_0 - (a_t + c_t)$, so $\Lambda = \max_t L_t = \sigma_0 - \min_t (a_t + c_t)$. Now $a_t + c_t = \max(t, A) + \max(u - t, C)$ is convex with increment -1 for $t < \tau_\wedge$, 0 on $[\tau_\wedge, \tau_\vee]$ and $+1$ for $t \geq \tau_\vee$, so its minimum is attained on $[\tau_\wedge, \tau_\vee]$. If $\tau_1 \leq \tau_2$ that range is region II, where $a_t + c_t = t + (u - t) = u$; if $\tau_2 < \tau_1$ it is region IV, where $a_t + c_t = A + C$. The two cases are distinguished by $\tau_1 \leq \tau_2 \iff A + C \leq u$, so in both $\min_t (a_t + c_t) = \max(u, A + C)$. $\square$

4 The radial form

Proposition 4.1 (radial form of the slice sum). *Define*

$$\beta(r) := \#\{t \in T : \delta_t \leq r \leq L_t\} \quad (r \in \mathbb{Z} - t_0), \quad (8)$$

and $\gamma(y) := \sum_{r \geq y} \beta(r)$. Then

$$G(s) = \gamma(|s - \sigma_0|) \quad \text{for all } s \in \mathbb{Z}. \quad (9)$$

Proof. First, $|s - \sigma_0|$ and δ_t, L_t all lie in the same coset of $\mathbb{Z}$: indeed $2\sigma_0 = u + B + D$ and $2t_0 = u + B - D$ differ by $2D$, so $\sigma_0 \equiv t_0 \pmod{1}$, while $\delta_t = |t - t_0| \in \mathbb{Z} - t_0$ and $L_t = n_t + (t - t_0) - 1 \in \mathbb{Z} - t_0$. So (9) is a statement on one lattice.

Fix t with $n_t, m_t \geq 1$ and write $x = s - \sigma_0$. By Proposition 3.2, Trap_t is supported on $[\sigma_0 - L_t, \sigma_0 + L_t]$, and the convolution of interval indicators of widths n_t, m_t is

$$\text{Trap}_t(s) = \min(s - (a_t + c_t) + 1, (b_t + d_t) - s + 1, n_t, m_t)_+ = \min(L_t + 1 - |x|, h_t)_+,$$

using $a_t + c_t = \sigma_0 - L_t$, $b_t + d_t = \sigma_0 + L_t$ and $\min(\alpha, \beta)_+ = \min(\alpha_+, \beta_+)$. Since $h_t = L_t + 1 - \delta_t$,

$$\min(L_t + 1 - |x|, h_t)_+ = (L_t - \max(|x|, \delta_t) + 1)_+ = \#\{r : \max(|x|, \delta_t) \leq r \leq L_t\},$$

the last count being over the lattice $\mathbb{Z} - t_0$. If instead $\min(n_t, m_t) \leq 0$ then $\text{Trap}_t \equiv 0$, and also $h_t \leq 0$, i.e. $L_t < \delta_t$, so no r satisfies $\delta_t \leq r \leq L_t$ and the count is 0 as well. Hence in all cases

$$\text{Trap}_t(s) = \#\{r : |x| \leq r, \delta_t \leq r \leq L_t\}.$$

Summing over $t \in T$ and exchanging the order of summation,

$$G(s) = \sum_{t \in T} \#\{r \geq |x| : \delta_t \leq r \leq L_t\} = \sum_{r \geq |x|} \beta(r) = \gamma(|x|). \quad \square$$

Proposition 4.2 (β is PF₂). *With Λ, α, ω as in Lemma 3.3 and $N = T_+ - T_- + 1 \geq 1$, put*

$$c(r) := \min \left(\underbrace{\min(T_+, t_0 + r, \omega - r)}_{\text{concave}} - \underbrace{\max(T_-, t_0 - r, \alpha + r)}_{\text{convex}} + 1, N(\Lambda + 1 - r) \right). \quad (10)$$

Then c is concave and $\mathbb{Z}$ -valued on $\mathbb{Z} - t_0$, and $\beta = c_+$. In particular β is PF₂ with finite support.

Proof. The row description. For $t \in \mathbb{Z}$ we have $\delta_t \leq r \iff |t - t_0| \leq r \iff t_0 - r \leq t \leq t_0 + r$. By Lemma 3.3, $L_t \geq r$ holds iff $\tilde{L}(t) \geq r$, which by (6) holds iff $r \leq \Lambda$ and $\alpha + r \leq t \leq \omega - r$. Hence for $r \leq \Lambda$

$$\beta(r) = \#\{t \in \mathbb{Z} : \max(T_-, t_0 - r, \alpha + r) \leq t \leq \min(T_+, t_0 + r, \omega - r)\}, \quad (11)$$

and $\beta(r) = 0$ for $r > \Lambda$.

Integrality. For $r \in \mathbb{Z} - t_0$ we have $t_0 + r \in \mathbb{Z}$ and $t_0 - r = 2t_0 - (t_0 + r) \in \mathbb{Z}$ since $2t_0 = u + B - D \in \mathbb{Z}$. Also $\Lambda \in \mathbb{Z} - t_0$ (it equals L_{τ_Λ} and $\tau_\Lambda \in \mathbb{Z}$), hence $\alpha = \tau_\Lambda - \Lambda \in \mathbb{Z} + t_0$ and $\omega = \tau_\vee + \Lambda \in \mathbb{Z} - t_0$, so $\alpha + r \in \mathbb{Z}$ and $\omega - r \in \mathbb{Z}$. Finally $T_\pm \in \mathbb{Z}$. Thus both the min and the max in (10) are integers, and the count in (11) equals $(\min(\dots) - \max(\dots) + 1)_+$ with no floors.

$\beta = c_+$. Write $c_1(r) = \min(T_+, t_0 + r, \omega - r) - \max(T_-, t_0 - r, \alpha + r) + 1$, so that $\beta(r) = c_1(r)_+$ for $r \leq \Lambda$. Since the min includes T_+ and the max includes T_- we have $c_1(r) \leq T_+ - T_- + 1 = N$ for every r . If $r \leq \Lambda$ then $\Lambda + 1 - r \geq 1$, so $N(\Lambda + 1 - r) \geq N \geq c_1(r)$ and $c(r) = c_1(r)$, whence $c(r)_+ = \beta(r)$. If $r > \Lambda$ then $r \geq \Lambda + 1$ (both lie in $\mathbb{Z} - t_0$), so $N(\Lambda + 1 - r) \leq 0$, hence $c(r) \leq 0$ and $c(r)_+ = 0 = \beta(r)$.

Concavity. A minimum of affine functions is concave and a maximum of affine functions is convex, so c_1 is concave; $N(\Lambda + 1 - r)$ is affine; a minimum of two concave functions is concave. Hence c is concave on $\mathbb{Z} - t_0$.

Conclusion. By [c1, Lem. 2.2] (`1em:trunc`, `lean-verified` as `TworowD4Kernel.DiscreteConcavity.PFtwo_p`) the positive part of a concave integer sequence has interval support and is log-concave; so β is PF₂. Its support is contained in $[0, \Lambda]$ — bounded below because $\delta_t \geq 0$ and above by Λ — hence finite. $\square$

Remark 4.3. The clamp $N(\Lambda + 1 - r)$ in (10) is *not* cosmetic. For $\Lambda < r \leq \Lambda + \frac{1}{2}|\tau_1 - \tau_2|$ the three-interval description in (11) is non-empty while $\{t : L_t \geq r\}$ is empty, because $\tilde{L}$ has a genuine flat piece and $\min(t - \alpha, \omega - t)$ exceeds Λ there. Dropping the clamp gives the wrong β in 7245 of 13 761 tested instances (§8).

5 The upper tail of a log-concave sequence

Lemma 5.1 (tail lemma). *Let $\beta : \mathbb{Z} \rightarrow [0, \infty)$ be log-concave with interval support and $\sum_r \beta(r) < \infty$. Then $\gamma(y) := \sum_{r \geq y} \beta(r)$ satisfies*

$$\gamma(y)^2 \geq \gamma(y-1)\gamma(y+1) \quad \text{for every } y \in \mathbb{Z},$$

γ is nonincreasing, and $\{\gamma > 0\}$ is an interval (a down-set).

Proof. Monotonicity and the shape of $\{\gamma > 0\}$ are immediate from $\beta \geq 0$: γ is nonincreasing, and $\gamma(y) > 0$ iff $y \leq r_1 := \max\{r : \beta(r) > 0\}$.

Fix y . From $\gamma(y-1) = \gamma(y) + \beta(y-1)$ and $\gamma(y+1) = \gamma(y) - \beta(y)$,

$$\gamma(y)^2 - \gamma(y-1)\gamma(y+1) = \beta(y-1)\beta(y) - \Delta\gamma(y), \quad \Delta := \beta(y-1) - \beta(y). \quad (12)$$

If $\Delta \leq 0$ the right side of (12) is ≥ 0 and we are done. So assume $\Delta > 0$, i.e. $\beta(y-1) > \beta(y) \geq 0$; we must show $\Delta\gamma(y) \leq \beta(y-1)\beta(y)$.

Case $\beta(y) = 0$. Then $\beta(y-1) > 0$, so $y-1$ lies in the support $[r_0, r_1]$, while $y \notin [r_0, r_1]$; since $y > y-1 \geq r_0$ this forces $y > r_1$, hence $\beta(r) = 0$ for all $r \geq y$ and $\gamma(y) = 0$. Both sides vanish.

Case $\beta(y) > 0$. Put $q = \beta(y)/\beta(y-1) \in (0, 1)$. Log-concavity gives $\beta(r+1)/\beta(r) \leq \beta(r)/\beta(r-1)$ whenever $\beta(r) > 0$; since the support is an interval, the ratios $\beta(r+1)/\beta(r)$ for $r_0 \leq r < r_1$ form a nonincreasing sequence. As $y-1 \geq r_0$ (because $\beta(y-1) > 0$) and the ratio at $r = y-1$ is q , we get $\beta(r+1)/\beta(r) \leq q$ for every r with $y-1 \leq r < r_1$. Telescoping,

$$\beta(y+k) \leq \beta(y)q^k \quad (k \geq 0),$$

(the inequality being $0 \leq \beta(y)q^k$ once $y+k > r_1$). Hence

$$\gamma(y) = \sum_{k \geq 0} \beta(y+k) \leq \beta(y) \sum_{k \geq 0} q^k = \frac{\beta(y)}{1-q} = \frac{\beta(y)\beta(y-1)}{\beta(y-1) - \beta(y)} = \frac{\beta(y-1)\beta(y)}{\Delta},$$

which is exactly $\Delta \gamma(y) \leq \beta(y-1)\beta(y)$. $\square$

Remark 5.2. Lemma 5.1 is the discrete form of the classical fact that a log-concave probability mass function has a log-concave survival function (equivalently, that the log-concave class is contained in the increasing-failure-rate class). I give the proof because it is three lines and because the hypothesis matters: *interval support is load-bearing but concavity is not*. The positive part of a concave sequence is more than enough; mere log-concavity suffices. Dropping log-concavity while keeping interval support destroys the conclusion in 82.6% of random tests (§8), so this is not a hypothesis carried along for safety.

Lemma 5.3 (radial lemma). *Let $\sigma_0 \in \frac{1}{2}\mathbb{Z}$ and let γ be nonnegative and nonincreasing on $\mathcal{Y} := \{|s - \sigma_0| : s \in \mathbb{Z}\}$ with $\gamma(y)^2 \geq \gamma(y-1)\gamma(y+1)$ for all $y \in \mathcal{Y}$ with $y \geq 1$, and with $\{\gamma > 0\}$ a down-set. Then $g(s) = \gamma(|s - \sigma_0|)$ is PF₂.*

Proof. Write $x = s - \sigma_0$ and note $\mathcal{Y} = \mathbb{Z}_{\geq 0}$ if $\sigma_0 \in \mathbb{Z}$ and $\mathcal{Y} = \frac{1}{2} + \mathbb{Z}_{\geq 0}$ otherwise.

- If $x \geq 1$ then $|x-1| = x-1$ and $|x+1| = x+1$, so the required inequality is $\gamma(x)^2 \geq \gamma(x-1)\gamma(x+1)$, which holds by hypothesis. If $x \leq -1$ the same argument applies to $-x$.
- If $x = 0$ (possible only when $\sigma_0 \in \mathbb{Z}$) then $g(s \pm 1) = \gamma(1)$ and $g(s) = \gamma(0) \geq \gamma(1) \geq 0$, so $g(s)^2 \geq \gamma(1)^2 = g(s-1)g(s+1)$.
- If $x = \frac{1}{2}$ (possible only when $\sigma_0 \notin \mathbb{Z}$) then $|x-1| = \frac{1}{2}$ and $|x+1| = \frac{3}{2}$, so the inequality reads $\gamma(\frac{1}{2})^2 \geq \gamma(\frac{1}{2})\gamma(\frac{3}{2})$, true since γ is nonincreasing and nonnegative. The case $x = -\frac{1}{2}$ is symmetric.

Finally $\{g > 0\} = \{s : |s - \sigma_0| \leq r_1\}$ where $r_1 = \max\{\gamma > 0\}$ (or $\{g > 0\} = \emptyset$), an interval of $\mathbb{Z}$. $\square$

6 Proof of Theorem 1.1

Proof of Theorem 1.1. Fix λ/μ with $m = 2$, a level $n \geq 2$ and a value of b . By [c1, Prop. 2.3], $k(a, b) = G(u + a)$ with G as in (2), $u = |\mu| + b$; so it suffices to prove $G \in \text{PF}_2$. If $T = \emptyset$ or every $t \in T$ has $\min(n_t, m_t) \leq 0$ then $G \equiv 0$ and there is nothing to prove.

Otherwise, let σ_0, t_0 be as in Proposition 3.2 and β, γ as in Proposition 4.1. By Proposition 4.2, β is a nonnegative log-concave sequence on $\mathbb{Z} - t_0$ with interval, finite support. By Lemma 5.1, γ is nonincreasing with $\{\gamma > 0\}$ a down-set and $\gamma(y)^2 \geq \gamma(y-1)\gamma(y+1)$ for all y . By Proposition 4.1, $G(s) = \gamma(|s - \sigma_0|)$, and the lattice $\{|s - \sigma_0| : s \in \mathbb{Z}\}$ is contained in $\mathbb{Z} - t_0$ because $\sigma_0 \equiv t_0 \pmod{1}$. Lemma 5.3 now gives $G \in \text{PF}_2$. $\square$

Corollary 6.1. *Corollary "single-region criterion" of [c1], which covered 79.9% of slices, is subsumed: condition (A) at $m = 2$ holds for every slice.*

7 Why (G) and concavity fail: an identity

The radial picture explains both negative results of this programme at once, and does so by an identity rather than by a census.

Proposition 7.1. *With β, γ as above, for every y*

$$2\gamma(y) - \gamma(y-1) - \gamma(y+1) = \beta(y) - \beta(y-1). \quad (13)$$

Hence G is concave at s with $|s - \sigma_0| = y \geq 1$ if and only if $\beta(y) \geq \beta(y-1)$; that is, G fails concavity exactly on the strictly decreasing part of β .

Proof. Substitute $\gamma(y-1) = \gamma(y) + \beta(y-1)$ and $\gamma(y+1) = \gamma(y) - \beta(y)$. □

So concavity of a slice sum holds iff its layer profile β is nondecreasing on its support, which it is not in general: β is c_+ for a concave c , and a concave sequence decreases eventually unless it is constant. This is the mechanism behind [c1, §"Concavity is false"]. Concretely, for the witness $k(\cdot, 2) = (1, 3, 6, 7, 6, 3, 1)$ at $n = 8$, $\mu = (0, 3)$, $\lambda = (4, 7)$ one has $\gamma = (7, 6, 3, 1)$ and $\beta = (1, 3, 2, 1)$, so $\beta(2) - \beta(1) = -1$ and concavity fails at $y = 2$, i.e. at the two points $\sigma_0 \pm 2$ — exactly the failure $2 \cdot 3 < 1 + 6$ recorded there.

Combining Proposition 7.1 with Lemma 2.2:

Corollary 7.2 ((G) is dead, with no residual scope). *Suppose two regions $R \neq R'$ are nonempty, $G_{R'}$ is constant and positive on an interval J , and some s with $s-1, s, s+1 \in J$ has $|s - \sigma_0^{(R)}| = y \geq 1$ where $\beta^{(R)}(y) < \beta^{(R)}(y-1)$. Then (G) fails for that pair. Region III contributes a constant G_{III} whenever T_{III} is a single t with $n_t = 1$ or $m_t = 1$, and region I likewise; so the configuration is not exceptional.*

This is the kind of reason the registry asks for: it has no quantifier left to shrink. (G) is not "unverified outside a range" and not "true with a side condition"; it is false for a structural reason that applies to any shape whose layer profile is non-constant beside a flat region, and the region decomposition is simply the wrong cut. The right cut is by *layer* r , not by region t : the regions slice the sum along t , where the family is a genuine antichain (registry node `lemma-T-layer-chain`: not a chain in 29.9% of instances), while the layers slice it along r , where the family is a nested chain of concentric intervals by Proposition 3.2.

Remark 7.3 (relation to Lemma T). This does *not* prove Lemma T of [c1, Conj. 2.6] in its stated generality (arbitrary concave 1-Lipschitz Φ and convex 1-Lipschitz Ψ). Concentricity, Proposition 3.2, uses the specific alignment $A = D - n + 1$, $C = B + 1$ of the cylindric data; for general Φ, Ψ the trapezoids need not share a centre. Lemma T remains open. What is now proved is the statement Lemma T was introduced to imply.

8 Verification

All code is in `scratch/1001-c2/`. Instruments were required to reproduce a known value or a known failure before any of their output was believed.

Instrument calibration.

- The region engine reproduces the known failure of $(*)$ between individual slice members at $n = 6$, $\mu = (0, 3)$, $\lambda = (4, 5)$, slice $S_\nu = 5$ ([c1, §2.1 item 1]): trapezoids $(1, 1, 1, 1, 1)$ on $[5, 9]$ for $t = 0$ and δ_7 for $t = 2$, failures at $s = 6, 8$. It also reproduces the individual PF_2 -ness of the four region sums.
- **Estimand check.** An *independent* instrument enumerates chains $\mu \prec \nu \prec \kappa \prec \lambda$ directly using the vendored primitives `is_shape`, `hstrip`, `size` of `proofs/code-q254/cyl.py` (the code behind the 2026-09-20 M-convexity paper), never touching a_t, b_t, c_t, d_t . Over $2 \leq n \leq 7$, $d \leq 11$: the symmetry control $k(a, b) = k(b, a)$ holds on 4908 of 4908 entries, and $G(u + a)$ agrees with $k(a, b)$ on all 3427 (shape, b) pairs. So the object being proved log-concave is $k(\cdot, b)$ and not a near-miss of it.
- **Refusal control for Lemma 5.1.** A first control — β log-concave but not truncated-concave — produced 0 failures in 5578 trials, i.e. it did not fire. That was a signal that the lemma was stronger than the concavity-based proof I first wrote, not that the instrument was fine; it led to the geometric-decay proof above. The corrected control — β with interval support but *not* log-concave — refuses 309 362 of 374 527 trials (82.6%), and small explicit probes ($\beta = (10, 1, 10)$, $(2, 1, 3)$, $(1, 0, 5)$) all fire. The instrument can say no.

Results.

- **Crossed-sum identities** (Lemma 3.1) and concentricity (Proposition 3.2), together with concavity and 1-Lipschitzness of L (Lemma 3.3): 0 violations over 292 290 slices, $2 \leq n \leq 12$, $d \leq 14$.
- **Radial form** (Proposition 4.1): $G = \gamma(|\cdot - \sigma_0|)$ on all 27 522 nonzero slices of the reduced sweep (0 mismatches). An earlier run reported 9616 mismatches; the defect was in my own tail accumulator, which built γ only down to $\text{minsupp } \beta$ instead of to the bottom of the lattice, so $\gamma(0)$ was never computed when $\beta(0) = 0$. Smallest exposing instance $n = 4$, $\mu = (0, 1)$, $\lambda = (0, 3)$, $b = 0$, where $G = (1, 1, 1)$ and the buggy reconstruction dropped the middle entry.
- β is PF_2 (Proposition 4.2): 0 failures on 27 522 instances; the explicit concave c of (10) reproduces β exactly on all 27 522 and is concave on the lattice in all 27 522. Control: deleting the clamp $N(\Lambda + 1 - r)$ makes the formula wrong in 7245 of 13 761 instances, smallest $n = 4$, $\mu = (0, 1)$, $\lambda = (0, 2)$, $b = 1$ — so the clamp is load-bearing. The clamp is *active* (ghost rows above Λ) in 14 490 of 27 522 instances.
- **Lemma 5.1:** 0 failures on 161 771 random PF_2 sequences and on an exhaustive enumeration of all 9654 PF_2 sequences in $\{0, \dots, 7\}^L$, $L \leq 6$; plus geometric and long flat sequences up to length 30.
- **Condition (A):** 0 failures over 201 147 nonempty slices, $2 \leq n \leq 11$, $d \leq 14$. Extended afterwards, by the direct trapezoid sum *without* using the radial shortcut, to $2 \leq n \leq 16$, $d \leq 20$ (89 366 nonzero slices, $\mu_1 = 0$ by translation invariance): 0 failures. The PF_2 test itself refuses $(1, 1, 2, 1, 1)$, so it can say no.
- **(G):** 92 failures over the same range; see Remark 2.4.

9 Gaps, and what $m \geq 3$ needs

1. **No gap in Theorem 1.1.** Every step is Lemma 3.1, Proposition 3.2, Lemma 3.3, Proposition 4.1, Proposition 4.2 (which cites [c1, Lem. 2.2], `lean-verified`), Lemma 5.1, Lemma 5.3. The one external input is [c1, Prop. 2.3], recorded `proved` in the registry.
2. **Lemma T is still open** and is now known to be strictly stronger than what (A) at $m = 2$ needs; see the remark at the end of §7.
3. $m \geq 3$. The proof used $m = 2$ twice, both times in Lemma 3.1: there are exactly two beads, so there are exactly two crossed sums, and each collapses. At general m the slice sum is $\sum_{\nu} *_{\mathbf{i}} \mathbf{1}_{[L_{\mathbf{i}}(\nu) - \nu_{\mathbf{i}}, R_{\mathbf{i}}(\nu) - \nu_{\mathbf{i}}]}$ over an $(m - 1)$ -dimensional slice of a box, so the relevant question is whether the summands are again concentric. The natural generalisation to test first is whether $\sum_{\mathbf{i}} (L_{\mathbf{i}}(\nu) + R_{\mathbf{i}}(\nu))$ is independent of ν on each slice. That is a one-line computation, and it is the next thing I would check.
4. **What this says about condition (B) and (L3).** Theorem 1.1 settles the $m = 2$ half of the $\ell = 3$ coupling gap (registry node `ell3-coupling-gap`); condition (B) (`conj-B-logsubmodular`) is untouched by this argument and remains `computed`.

10 Summary

The brief asked me to prove (G) and warned me, correctly, to extend the range first. The range extension refuted (G) in the first computation of the session, and the refutation was available all along inside the paper that posed it: against a flat partner the pairwise inequality *is* concavity, and that paper proves concavity false. The repair was not a weaker version of (G) but a different decomposition. Cutting the slice sum by layer instead of by region turns an antichain into a nested chain of concentric intervals, and then log-concavity is the classical statement that the tail of a log-concave sequence is log-concave. Condition (A) at $m = 2$ is proved.